

Ross Kelso

Edinburgh, Scotland

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Software developer with 5 years' experience in full-stack development and a first class MSci in Computer Science from St Andrews.

Specialized in rapid development of data analysis and visualisation systems using `c#`, `python`, and `javascript`, with a background in machine learning, computer vision, and industrial software.

Technical Skills

Languages & Frameworks: `c#`, `python`, `javascript`, `typescript`, `.net`, `angular`, `react`, `sql`, `rust`

Technologies: `elasticsearch`, `git`, `azure`, `aws`, `docker`, `kubernetes`, `tensorflow`, `nodejs`, `windows`, `linux`, `ci/cd`

Concepts: Machine Learning, Data Engineering, Cloud Architecture, Computer Vision

Work Experience

Cintra HR & Payroll - Software Developer (since June 2025)

Creating HR & payroll management software.

`c#` `typescript` `angular`

- Through rigorous automated testing, identified and fixed longstanding architectural issues with existing validation middleware resulting in improved validation accuracy and data consistency.
- Applied source generation to ensure type safe and performant interoperability between new and legacy systems.
- Delivered new pensions functionality ahead of schedule while working in a cross-functional team.
- Communicated technical and domain problems clearly and effectively.
- Designed and implemented performance testing methodology to assess API throughput at scale, achieving a 5x increase in maximum throughput.

Intelligent Plant - Software Developer (2021-2025), Intern (University Holidays 2017-2021)

Creating data storage, analysis and visualisation solutions for the energy industry.

Lead developer for the [Industrial App Store API Python Client](#) `python` `pandas` `django` `tensorflow`

- Designed and implemented a Python SDK to simplify access to the Industrial App Store API for data science and ML use cases.
- Wrote comprehensive documentation and example machine learning use cases for ease of adoption by industry partners.

Lead developer for [Sequence of Events Explorer](#) `c#` `javascript` `html` `css` `elasticsearch` `ci`

- Developed the application from inception to maturity, including design, development, deployment, and maintenance resulting in it being adopted by multiple major industrial clients.
- Enhanced user experience by developing an intuitive, and powerful frontend for advanced data queries.
- Improved code quality and maintainability by managing developers through rigorous code reviews and enforcing best practices.

Developer for [Alarm Analysis](#) and Intelligent Plant Historian `c#` `elasticsearch`

- Created a novel ML approach for parsing non-standard semi-structured alarm data.
- Developed and maintained a backend for performing complex data aggregations.
- Worked with clients directly to facilitate on-site installations.
- Upgraded our database technology, doubling the size of a query that can be performed in 1s and increasing the size of a query that can be performed in 5s by an order of magnitude.

Developer for [Gestalt \(PnID & Trend\)](#) `c#` `sql` `javascript`

- Led the project to decompose our monolithic API into separate application and data components, resulting in smoother deployments for Gestalt and our data API.
- Designed and implemented a database model to replace an existing file system dependency paving the way for more reliable and scalable deployment.

Experimental projects involving the application of AI and large language models to our software `python` `docker` `tensorflow` `ollama`

- Deployed internal AI inference APIs.
- Researched and developed proof-of-concept applications integrating LLMs for industrial automation.

Lead developer for [Power BI Industrial Connector](#) `c#` `power query m` `react`

- Developed a high-performance data connector to enable seamless industrial data access in Power BI.
- Re-architected the connector to remove tech debt and improve UX.

Education

University of St Andrews - Computer Science MSci First Class (2016-2021)

Deans' List, for academic excellence 2017/18, 2018/19, 2019/20

Masters Thesis

[Synthetic GPS Traces for Privacy Protection](#) `python` `tensorflow` `pandas` `numpy`

- Applied Markov Models and generative AI to synthesize realistic GPS traces.
- Evaluated model accuracy and effectiveness in preserving statistical properties of real-world movement data.

Senior Honours Project

[Computer Vision in Cue Sports](#) `opencv` `python`

- Designed an algorithmic computer vision system to analyze pool table positions without the need for markers.
- Implemented perspective transformation and object detection techniques for accurate ball position detection.

Junior Honours Project

Group software engineering project to create an encrypted wireless network of embedded devices `c` `nodejs` `python` `git`

- Applied low-level networking principles to develop a secure, distributed communication protocol.
- Implemented encryption mechanisms for secure data transmission.
- Worked in an Agile environment, utilizing SCRUM methodologies for iterative development and collaboration.

Other Modules

- Algorithms and data structures `java`

- Artificial intelligence and machine learning `python` `java`
- Information Visualisation `python`
- Operating systems `c`
- Advanced programming projects `haskell` `python`

Personal Interests

Game Design `rust` `bevy` `godot`

Competed in the [GMTK Game Jam 2025](#) using `godot`. Implemented a [Tetris clone](#) in `rust` using `SDL2` including a simple software synthesizer to play the classic music. Implemented a [Flappy Bird clone](#) in `rust` using `bevy` game engine.

Internet of Things `c` `python` `nodejs`

- Created a network of micro-controllers that recorded temperature data around my flat to an Intelligent Plant Historian
- Created a website showing whether communal items in my flat had run out, updated by an ESP-32 stuck to the fridge.

Starfinder/D&D Game Master (since 2024)

- Coordinated and led a group of players through an immersive and engaging story, ensuring a cohesive and enjoyable experience for all participants.
- Designed challenging scenarios that required players to think critically and strategically to overcome.
- Improvised and responded to unexpected player choices or rule interpretations, using creative problem-solving and diplomacy to resolve conflicts and maintain a positive game environment.

Pool Society Vice-president, Social Rep, Publicity and Fundraising Rep (2017-2021)

- Managed day-to-day operations, including equipment maintenance, weekly sessions and recruiting new members.
- Promoted the society through social media channels and organized fundraisers, including live streams of semesterly Finals Day events.
- Represented the university at several pool competitions, including 5 BUCS tournaments.